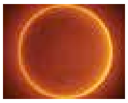
First woman on the Moon
NASA is preparing to send the very first woman as well as the next man to the moon. **https://dgit.in/aug19-17**



NASA shares a breathtaking photo
NASA shared an astounding shot of the ISS passing right in front of the sun, shot by Rainee Colacurcio. **https://dgit.in/aug19-18**

from getting stripped away by the action of solar wind. The Mars Express and Maven missions have directly observed the action of the solar wind on the tenuous Martian atmosphere. A proposed idea to preserve the atmosphere, is to introduce a magnetic shield at the Mars L1 Lagrange point, between Mars and the Sun. The shield would act as an artificial magnetosphere, and protect Mars from the effects of solar wind. The researchers expect such a shield to increase the temperature by about four degrees Celsius, melt the carbon dioxide in the polar regions, further increasing the atmospheric density and temperature. As a consequence of the released carbon dioxide working as a greenhouse gas, the temperatures would rise further, melting the ice trapped in the polar regions. Other studies are not so optimistic.

In 2018, NASA-funded researchers showed that there simply was not enough carbon dioxide in Mars that could be released into the atmosphere, to warm the planet enough so that liquid water could exist on the surface. The study took into account over 20 years of observations by instruments on Mars and in Martian orbit. Christopher Edwards, the co-author of the study, explains, "These data have provided substantial new information on the history of easily vaporized (volatile) materials like CO_2 and H_2O on the planet, the abundance of volatiles locked up on and below the surface, and the loss of gas from the atmosphere to space". Even if all the resources on Mars would be used up, the resulting atmosphere would only be seven percent as dense as the Earth, and not enough to warm the planet sufficiently for say, growing crops. The study also showed that most of the carbon dioxide that was available on Mars, was actually trapped in locations that are hard to reach.

Bruce Jakosky, the lead author of the study said, "Our results suggest that there isn't enough CO_2 remaining on Mars to provide significant greenhouse warming if the gas were to be put into the atmosphere; in addition, most of the CO_2 gas is not accessible and could not be readily mobilised. As a result, terraforming Mars is not possible using present-day technology."

TERRAFORMING IN PATCHES

Now, researchers from Harvard University have proposed an approach that could make portions of Mars habitable, and allow any future manned-missions to grow food from the regolith. Instead of terraforming the whole planet, regions could be terraformed. The approach uses existing technology, and is feasible right now. The idea is to introduce silica aerogel, directly into the regolith, or the dry Martian soil. Silica aerogel is an incredibly light material, made up of 99 per cent air. It is a bit like styrofoam, and can play the same role on Mars as the greenhouse gases do on Earth. It can trap heat, and make the surface warm enough for liquid water to exist on the surface, allowing for the cultivation of crops.

Credit: NASA/JPL-Caltech

Silica aerogel

The aerogel is translucent, allowing essential sunlight for photosynthesis to pass through, and blocking harmful ultraviolet radiation at the same time. A shield of silica aerogel between two to three centimeters in thickness can permanently increase surface temperatures to allow liquid water to exist on the surface. The approach does not require an energy source for producing heat, and the maintenance is minimal, as there are no complicated structures and moving parts. Additionally, the same material can be incorporated into greenhouses and habitats for any future pioneers that visit Mars. The approach can potentially allow manned exploration of the Martian surface to be more expansive, and cover more regions. Using silica aerogel to reduce the surface temperatures of Mars in local islands of habitability, will be particularly effective in the middle latitudes, in places where there is water ice trapped near the surface.

Robin Wordsworth, Assistant Professor of Environmental Science and Engineering at Harvard explains, "this regional approach to making Mars habitable is much more achievable than global atmospheric modification. Unlike the previous ideas to make Mars habitable, this is something that can be developed and tested systematically with materials and technology we already have."

The researchers were inspired by a process that takes place naturally on the red planet. Just like the Earth, Mars has polar ice caps as well. While the polar ice caps on Earth are made up of water, on Mars, they are made up of a combination of water and carbon dioxide. The carbon dioxide ice provides a solid state greenhouse gas effect, trapping the heat just like the gaseous form. The ice traps heat beneath the surface, eventually causing the ice to rupture in black spots, where gas and soil escape from beneath the surface. The silica aerogel provides a solid state greenhouse gas effect just like the carbon dioxide ice. It is a good insulator, and is translucent enough to allow sunlight to pass through it.

The experiment is a proof of concept, and there are still some challenges to overcome. The climate model used for the simulation showed that it would take two Martian years or four Earth years to produce a permanent layer of subsurface liquid water. There is also the question of getting the aerogel to Mars. It would require shipping large amounts of the material, or finding a way to manufacture it in situ. The researchers want to test the approach in the field, such as the McMurdo Dry Valleys in Antarctica or the Atacama desert in Chile, both of which have deserts with little moisture.